

# The Loyola Index Family: Bounds, Parameter Geometry and Sensitivity Analysis

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## Abstract

The two-parameter Gutman–Milovanović family  $M_{\alpha,\beta}(G) = \sum_{uv \in E(G)} (d_u d_v)^\alpha (d_u + d_v)^\beta$  unifies the Zagreb, Randić, sum-connectivity and related degree-based indices. We study its normalized degree-imbalance exponential extension, the *Loyola index family*

$$LO(G; \alpha, \beta, \gamma) = \sum_{uv \in E(G)} (d_u d_v)^\alpha (d_u + d_v)^\beta \exp\left(\gamma \frac{|d_u - d_v|}{d_u + d_v}\right),$$

with  $LO(G; \alpha, \beta, 0) = M_{\alpha,\beta}(G)$  identically.

We prove that the imbalance direction is affinely independent of the two parent directions on the degree-pair set of graphs with maximum degree at least 3, that  $\log LO$  is globally convex in  $(\alpha, \beta, \gamma)$  with gradient and Hessian given by the mean and covariance of the edge-feature vector under an induced edge measure (the standard log-sum-exp property), that the family factorises into a size coordinate and a shape (degree-ratio) coordinate, and that on the resulting ratio plane the pure- $\gamma$  points separate exactly the histograms of

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edge imbalances (for algebraic  $\gamma \neq 0$ ); we also bound the number of  $\gamma$ -crossings of two graphs and characterise per-graph identifiability. On molecular graphs the entire family is confined to the finite-dimensional space of edge-degree-pair counts; the pure- $\gamma$  points attain this ceiling on trees up to order 12 (order 15 for  $\Delta \leq 4$ ) and provably not beyond. In nested leave-one-out comparisons on five octane properties and a pre-registered test on the 35 nonanes, freeing  $\gamma$  appeared to improve prediction of the vaporization enthalpy, but this gain vanishes once the parent family is searched over a box that does not truncate it; for this property the untuned point  $LO(0, 0, 1)$ , essentially the irregularity index *IRLA*, matches the tuned models, and on the nonanes ridge regression on all degree-pair counts predicts better. Hence  $\gamma$  is presented as a structurally interpretable, parsimonious direction rather than a predictive upgrade.

## 1 Introduction

The literature on degree-based topological indices in chemical graph theory has grown substantially since the seminal Wiener index [53], the Randić index [46], and the Zagreb indices of Gutman and Trinajstić [28]. Among the many parametric extensions are the general Randić index  $R_\alpha$  [8], the general sum-connectivity index  $\chi_\alpha$  [56], the forgotten index  $F$  revisited by Furtula and Gutman [16], the geometric-arithmetic index  $GA$  [51], the Albertson irregularity index [2] and the inverse  $\sigma$ -index problem [27].

A geometric strand was opened by the Sombor index family [21, 22], followed by work on its bounds [37], its relations to classical degree-based indices [52], and its extremal values over molecular trees and related classes [1, 11, 13]. Its variants include the mean [36], elliptic [25], hyperbolic [7], diminished [39, 40] and KG-Sombor [33] indices (see the survey [23]). Such descriptors are increasingly used in quantitative structure-property relationship (QSPR) and spectral studies [5, 31], and in matrix, coindex and simplicial formulations [20, 48, 57] and finite-dimensional linear frameworks [44]; extremal results on bond-incident-degree (BID) indices in general are surveyed in [3].

The natural two-parameter unification of the product-based and sum-based degree indices is the *Gutman-Milovanović family*, introduced by Gutman, Milovanović and Milovanović [26] and studied

intensively in the recent bounds and QSPR literature [19, 38, 42]:  $M_{\alpha,\beta}(G) = \sum_{uv \in E(G)} (d_u d_v)^\alpha (d_u + d_v)^\beta$ ,  $\alpha, \beta \in \mathbb{R}$ . Its special cases (most of them catalogued in [38]) include the first and second Zagreb indices  $M_{0,1} = M_1$  and  $M_{1,0} = M_2$ , the hyper-Zagreb index  $M_{0,2} = HM$  [49], the modified second Zagreb index  $M_{-1,0} = {}^m M_2$ , the Randić index  $M_{-1/2,0} = R$ , the sum-connectivity index  $M_{0,-1/2} = \chi$  [55], the harmonic index  $2M_{0,-1} = H$ , the inverse sum indeg index  $M_{1,-1} = ISI$ , the geometric-arithmetic index  $2M_{1/2,-1} = GA$ , the arithmetic-geometric index  $\frac{1}{2}M_{-1/2,1} = AG$ , and, as parameter axes, the general Randić family  $R_\alpha = M_{\alpha,0}$  [8] and general sum-connectivity family  $\chi_\beta = M_{0,\beta}$  [56]. Parameter-optimized versions of this family and of related generalized descriptors have already been fitted to octane properties [4]. We claim no novelty for this backbone, for its reduction catalogue, or for its generic tunability: those belong to the Gutman–Milovanović literature.

In this paper we study a three-parameter extension of  $M_{\alpha,\beta}$ , the *Loyola index family*, obtained by attaching a separately parameterized exponential factor in the normalized degree imbalance  $q_{uv} = |d_u - d_v|/(d_u + d_v)$  of each edge:  $LO(G; \alpha, \beta, \gamma) = \sum_{uv} (d_u d_v)^\alpha (d_u + d_v)^\beta e^{\gamma q_{uv}}$ , so that  $LO(G; \alpha, \beta, 0) = M_{\alpha,\beta}(G)$  identically. Positive  $\gamma$  exponentially upweights degree-imbalanced edges, negative  $\gamma$  downweights them, and on regular graphs  $\gamma$  has no effect. The questions this paper answers are therefore about  $\gamma$  alone: is the imbalance direction mathematically independent of the parent parameters, what analytic structure does it carry, and does it add measurable structural or predictive information beyond an optimized Gutman–Milovanović parent?

Writing the summand as  $\exp(\alpha \ln(d_u d_v) + \beta \ln(d_u + d_v) + \gamma q_{uv})$  shows that the family is a three-parameter subfamily of the exponential vertex-degree-based indices introduced by Rada [43] and studied in extremal settings by Sigarreta [50]; general sufficient conditions for tree-extremal behaviour of such exponential edge-weight sums are known [18] (see also [10, 12]), bounds for general edge-weight functions over graphs of given order and size are given in [29], and no extremal-ordering novelty is claimed here. The published one-parameter exponential indices (e.g. the exponential geometric–arithmetic [14] and exponential inverse symmetric division

deg [15] indices) place their parameter on a single classical generator inside the exponent; the scoped contribution of this family is the multi-parameter *linear combination* in the exponent in which the two Gutman–Milovanović basis terms act as the factors  $(d_u d_v)^\alpha (d_u + d_v)^\beta$  while the imbalance coordinate carries its own parameter  $\gamma$ . The imbalance ratio itself appears in many published indices: the diminished Sombor index, with edge weight  $\sqrt{(1 + q_{uv}^2)/2}$  [40]; the irregularity index  $IRLA(G) = 2 \sum_{uv} q_{uv}$  [47], of which the pure- $\gamma$  point  $LO(G; 0, 0, \gamma) = \sum_{uv} e^{\gamma q_{uv}}$  is the exponential index in the sense of [43]; degree-ratio indices [24, 41]; and the parent family itself, since  $1 - q_{uv}^2 = 4d_u d_v / (d_u + d_v)^2$  gives  $M_{\alpha, -2\alpha}(G) = 4^{-\alpha} \sum_{uv} (1 - q_{uv}^2)^\alpha$  (the variable geometric–arithmetic direction). The parent family is therefore not blind to imbalance; what is new is a coordinate *linear* in  $q_{uv}$  in the exponent, which, unlike  $\ln(1 - q_{uv}^2)$ , is not an affine function of  $\ln(d_u d_v)$  and  $\ln(d_u + d_v)$  (Proposition 2). To our knowledge, and within the scope of the literature search documented in the reproducibility package, no published index couples it multiplicatively, with its own parameter, to the Gutman–Milovanović family. Since  $0 \leq q_{uv} < 1$ , the factor lies in  $[1, e^\gamma)$  for  $\gamma > 0$ , equals 1 for  $\gamma = 0$ , lies in  $(e^\gamma, 1]$  for  $\gamma < 0$ , and reduces to 1 on regular graphs. The exponential form is not derived from a physical model; it is chosen because it makes the family log-linear in its parameters, so that  $\gamma$  acts multiplicatively and only on irregularity, and the whole family is an exponential family in  $(\alpha, \beta, \gamma)$ , which is what makes the convexity, identifiability and discrimination results of Section 3 available.

On hydrogen-suppressed molecular graphs with  $\Delta \leq 4$ , every BID index is a linear functional of the ten edge-degree-pair counts  $m_{ij}(G) = |\{uv \in E(G) : \{d_u, d_v\} = \{i, j\}\}|$ ,  $1 \leq i \leq j \leq 4$  (cf. the linear framework of Rada [44]), so the Loyola family lies inside a fixed ten-dimensional descriptor space, however its parameters are chosen.

*Scope of the contribution.* Four questions should be answered at the outset. First, is the family sufficiently novel, given  $IRLA$  and the existing exponential vertex-degree-based indices? The edge invariant  $q_{uv}$  is classical, and the pure- $\gamma$  point  $LO(0, 0, \gamma)$  is simply the exponential index, in the sense of [43], of the  $IRLA$  edge weights. What is new is not a new edge invariant but its embedding as a free, log-linear coordinate inside

the Gutman–Milovanović family, and the exact description of the resulting three-parameter structure: a size–shape factorisation, a ratio plane on which discrimination is characterised exactly, and per-graph identifiability. Second, several of the mathematical results are applications of standard tools: the convexity of log-sum-exp functions, Laguerre’s rule of signs, the Lindemann–Weierstrass theorem and the minimality theory of exponential families. We claim no novelty for these tools; the contribution is their application to this family and the structural characterisation that results (the exact  $q$ -histogram ceiling of the pure- $\gamma$  line, colliding tree families at every order from 13, a bound on  $\gamma$ -crossings, and the classification of scale-invariant directions). Third, why use a three-parameter family if *IRLA* performs similarly? Not for predictive superiority, which we do not find. The reasons are interpretability within the parent family (every member is a deformation of a classical Gutman–Milovanović index, recovered at  $\gamma = 0$ ); continuous structural deformation (size and shape are controlled separately, and  $\gamma$  moves continuously from the parent towards imbalance-weighted indices); ratio-plane geometry (the pure- $\gamma$  line,  $GA$  and  $AG$  lie in one plane on which their discriminating power is described exactly); and analytic tractability (log-linearity gives convexity, closed-form derivatives such as  $\partial_\gamma \log LO = \mathbb{E}_\theta[q]$ , and an exact identifiability theory). *IRLA* is, to first order in  $\gamma$ , a single slice of this structure. Fourth, the QSPR datasets are small: 18 octanes and 35 nonanes. We therefore present the chemometric component as diagnostic evidence about what imbalance information does and does not add to the parent family, not as proof of chemical predictive utility.

In sum, the contributions are: (i) independence and per-graph identifiability of the imbalance direction (Propositions 2, 6); (ii) log-convexity with mean/covariance derivatives (Theorem 3); (iii) the ratio-plane theorem (Theorem 4), with tree collisions at every order from 13 (from 16 for  $\Delta \leq 4$ ), and the crossing bound (Proposition 5); (iv) closed forms,  $\gamma \neq 0$  bounds, the degree-pair ceiling (Proposition 14) and structure sensitivity; and (v) a QSPR diagnostic (Section 6) in which freeing  $\gamma$  gives no consistent gain once the parent search box does not truncate the parent. No universal ranking or predictive-superiority claim is made; the family is

offered as an analytically transparent, parsimonious third direction on the Gutman–Milovanović parameter space.

## 2 Preliminaries

### 2.1 Notation

Let  $G = (V, E)$  be a finite simple connected graph of order  $n = |V|$  and size  $m = |E|$ . For a vertex  $v \in V$ , the degree  $d_v = d_G(v)$  counts the edges incident to  $v$ . The maximum and minimum degrees are denoted  $\Delta = \Delta(G)$  and  $\delta = \delta(G)$ . In the chemometric sections we work with hydrogen-suppressed molecular graphs, where the assumption  $\Delta \leq 4$  holds for alkane carbon skeletons (all graphs analysed here) and for most organic molecules, but not universally: hypervalent centres such as phosphorus in  $\text{PCl}_5$  (degree 5) or sulfur in pentafluorosulfanyl groups (degree 6) exceed it.

The classical graph families used are the complete graph  $K_n$ , the cycle  $C_n$ , the path  $P_n$ , the star  $S_n$ , the complete bipartite graph  $K_{p,q}$ , the hypercube  $Q_k$  ( $2^k$  vertices,  $k \cdot 2^{k-1}$  edges,  $k$ -regular), the wheel  $W_n$  (an  $(n-1)$ -cycle plus a universal vertex), the friendship graph  $F_p$  ( $p$  triangles sharing a common vertex), and the  $r$ -regular graphs. For an edge  $e = uv \in E(G)$  we write  $(d_u, d_v)$  for its *degree pair*, understood as an unordered multiset  $\{d_u, d_v\}$ .

### 2.2 Definition

**Definition 1** (Loyola index family). Let  $G = (V, E)$  be a simple connected graph. For  $\alpha, \beta, \gamma \in \mathbb{R}$ , the *Loyola index family* is defined as,

$$LO(G; \alpha, \beta, \gamma) = \sum_{uv \in E(G)} (d_u d_v)^\alpha (d_u + d_v)^\beta \exp\left(\gamma \frac{|d_u - d_v|}{d_u + d_v}\right) \quad (1)$$

Equation (1) is well defined for all  $\alpha, \beta \in \mathbb{R}$  on graphs without isolated vertices (so that  $d_u, d_v \geq 1$  on every edge). The expression is invariant under permutations of  $(u, v)$  because  $d_u d_v = d_v d_u$ ,  $d_u + d_v = d_v + d_u$ , and

$|d_u - d_v| = |d_v - d_u|$ ; hence  $LO(G; \alpha, \beta, \gamma)$  depends only on the multiset of unordered edge degree pairs of  $G$ . When  $\gamma = 0$  the family reduces identically to the Gutman–Milovanović family [26, 38],

$$LO(G; \alpha, \beta, 0) = M_{\alpha, \beta}(G) = \sum_{uv \in E(G)} (d_u d_v)^\alpha (d_u + d_v)^\beta,$$

whose classical special cases are catalogued in the Introduction. Throughout,  $q_{uv} = \text{imb}(d_u, d_v)$ , where  $\text{imb}(x, y) = |x - y|/(x + y)$ , denotes the normalized degree imbalance of the edge  $uv$ ; positive  $\gamma$  exponentially up-weights degree-imbalanced edges, negative  $\gamma$  downweights them, and on regular graphs every  $q_{uv} = 0$  so that  $\gamma$  has no effect and  $LO(G; \alpha, \beta, \gamma) = M_{\alpha, \beta}(G)$  for all  $\gamma$ .

## 2.3 Closed-form values on standard graphs

**Proposition 1.** *Let  $\alpha, \beta, \gamma \in \mathbb{R}$ . Then:*

1.  $LO(K_n; \alpha, \beta, \gamma) = \frac{n(n-1)}{2} 2^\beta (n-1)^{2\alpha+\beta}$  for  $n \geq 2$ .

2.  $LO(C_n; \alpha, \beta, \gamma) = n \cdot 4^{\alpha+\beta}$  for  $n \geq 3$ .

3.  $LO(P_n; \alpha, \beta, \gamma) = \begin{cases} 2^\beta, & n = 2, \\ 2 \cdot 2^\alpha 3^\beta e^{\gamma/3}, & n = 3, \\ 2 \cdot 2^\alpha 3^\beta e^{\gamma/3} + (n-3) \cdot 4^{\alpha+\beta}, & n \geq 4. \end{cases}$

4.  $LO(S_n; \alpha, \beta, \gamma) = (n-1)(n-1)^\alpha n^\beta \exp\left(\gamma \frac{n-2}{n}\right)$  for  $n \geq 2$ .

5.  $LO(K_{p,q}; \alpha, \beta, \gamma) = pq(pq)^\alpha (p+q)^\beta \exp\left(\gamma \frac{|p-q|}{p+q}\right)$ .

6.  $LO(Q_k; \alpha, \beta, \gamma) = k \cdot 2^{k-1} \cdot 2^\beta k^{2\alpha+\beta}$  for  $k \geq 1$ .

7. For  $n \geq 5$ ,  $W_n$  has  $n-1$  spoke edges with degree pair  $(3, n-1)$  and  $n-1$  rim edges with degree pair  $(3, 3)$ , so

$$LO(W_n; \alpha, \beta, \gamma) = (n-1) \left[ 3^\alpha (n-1)^\alpha (n+2)^\beta \exp\left(\gamma \frac{n-4}{n+2}\right) + 9^\alpha 6^\beta \right].$$

8. For  $p \geq 1$ ,  $F_p$  has  $2p$  spoke edges  $(2, 2p)$  and  $p$  outer edges  $(2, 2)$ , so

$$LO(F_p; \alpha, \beta, \gamma) = p \cdot 4^{\alpha+\beta} + 2p(4p)^\alpha(2p+2)^\beta \exp\left(\gamma \frac{p-1}{p+1}\right).$$

9. For an  $r$ -regular graph  $G$  with  $m$  edges,  $LO(G; \alpha, \beta, \gamma) = m \cdot 2^\beta r^{2\alpha+\beta}$ .

*Proof.* Each case follows by enumerating the edge degree pairs and applying Definition 1. In  $K_n$ ,  $C_n$ ,  $Q_k$  and  $r$ -regular graphs every edge has  $d_u = d_v = r$ , so  $\text{imb} = 0$  and  $(d_u d_v)^\alpha (d_u + d_v)^\beta = 2^\beta r^{2\alpha+\beta}$ , which gives items (1), (2), (6) and (9). All  $n-1$  edges of  $S_n$  have degree pair  $(1, n-1)$  with  $\text{imb} = (n-2)/n$  (item (4)); all  $pq$  edges of  $K_{p,q}$  have pair  $(p, q)$  (item (5));  $P_n$ ,  $n \geq 4$ , has two pendant edges  $(1, 2)$  and  $n-3$  interior edges  $(2, 2)$ , and the cases  $n = 2, 3$  are immediate (item (3)); items (7) and (8) follow by summing over the two edge-degree-pair classes of  $W_n$  and  $F_p$ . ■

### 3 Parameter geometry and the role of the imbalance parameter

This section collects the structural results. Throughout,  $z_{uv} = (\ln(d_u d_v), \ln(d_u + d_v), q_{uv})$  is the feature vector of the edge  $uv$  and  $\theta = (\alpha, \beta, \gamma)$ , so that  $LO_G(\theta) = \sum_{uv \in E(G)} e^{\theta^\top z_{uv}}$ . The independence result below is a short computation (almost any edge feature that is not log-linear in the degrees would pass the same test).

**Proposition 2** (Affine independence of the imbalance direction). *Let  $\Delta \geq 3$  and let  $P_\Delta = \{(i, j) : 1 \leq i \leq j \leq \Delta\}$  be the admissible degree pairs. Then the augmented feature vectors  $(1, \ln(ij), \ln(i+j), q_{ij})$ ,  $(i, j) \in P_\Delta$ , span  $\mathbb{R}^4$ . Consequently, if two parameter triples  $\theta, \theta'$  satisfy  $e^{\theta^\top z} = c e^{\theta'^\top z}$  for some constant  $c > 0$  and all  $(i, j) \in P_\Delta$ , then  $\theta = \theta'$  and  $c = 1$ ; in particular no choice of  $(\alpha', \beta')$  and scaling constant reproduces the action of  $\gamma \neq 0$  on all admissible degree pairs. For  $\Delta = 2$  the span has dimension 3, and for  $\Delta = 1$  (the single pair  $(1, 1)$ , augmented vector  $(1, 0, \ln 2, 0)$ ) it has dimension 1; in both cases the conclusion fails.*

*Proof.* It suffices to show that  $q_{ij}$  is not an affine function of  $(\ln(ij), \ln(i+j))$  on  $P_\Delta$ ; the four pairs  $(1, 1), (2, 2), (1, 2), (1, 3)$ , all admissible once  $\Delta \geq$

3, witness this. Suppose  $q_{ij} = a + b \ln(ij) + c \ln(i + j)$  on these pairs. From  $(1, 1)$  and  $(2, 2)$ , where  $q = 0$ :  $a + c \ln 2 = 0$  and  $a + (2b + 2c) \ln 2 = 0$ , giving  $a = -c \ln 2$  and  $b = -c/2$ . Substituting into the  $(1, 2)$  and  $(1, 3)$  equations gives  $\frac{1}{3} = c(\ln 3 - \frac{3}{2} \ln 2)$  and  $\frac{1}{2} = c(\ln 2 - \frac{1}{2} \ln 3)$ ; in particular  $c \neq 0$ . Three times the first minus twice the second gives  $0 = c(4 \ln 3 - \frac{13}{2} \ln 2)$ , hence  $8 \ln 3 = 13 \ln 2$ , i.e.  $3^8 = 2^{13}$ , which is false ( $6561 \neq 8192$ ). Hence the augmented vectors have rank 4. If  $e^{\theta^\top z} = c e^{\theta'^\top z}$  on  $P_\Delta$ , then  $(\theta - \theta')^\top z_{ij} = \ln c$  for all admissible pairs, i.e.  $(\ln c, \theta' - \theta)$  is orthogonal to every augmented vector, so  $\theta = \theta'$  and  $c = 1$ . For  $\Delta = 2$  only the pairs  $(1, 1), (1, 2), (2, 2)$  exist and a direct computation shows the augmented vectors have rank 3; for  $\Delta = 1$  the single augmented vector has rank 1.  $\blacksquare$

*Remark 1* (Independence versus per-graph identifiability). Proposition 2 concerns the edge-weight *function* on the full admissible pair set, not every individual graph: on a single graph  $G$  the parameters are identifiable from the *function*  $\theta \mapsto LO(G; \theta)$  (never from a single value of it, which is one scalar), up to the scaling noted above, only when the degree pairs realised by  $G$  carry four affinely independent augmented feature vectors. On a regular graph nothing about  $\theta$  is identifiable beyond one linear functional; if the edges of  $G$  realise at most three affinely independent augmented vectors, some direction of  $\theta$  — possibly the  $\gamma$  direction — is unidentifiable from  $LO(G; \cdot)$  alone. The proposition guarantees that no such degeneracy is forced on the whole molecular class  $\Delta \geq 3$ : the family separates parameters as a family of descriptors, even though particular graphs may not.

The next result is the standard convexity of a log-sum-exp (log-partition) function, whose gradient and Hessian are the mean and covariance of the exponent’s feature vector under the induced weights; we state it for  $LO$  because these moments have a direct interpretation here and because strictness depends on the degree pairs that  $G$  realises.

**Theorem 3** (Global log-convexity). *Fix a graph  $G$  with  $m \geq 1$  edges and define  $F_G(\theta) = \log LO_G(\theta)$  on  $\mathbb{R}^3$ . Let  $\mathbb{E}_\theta$  and  $\text{Cov}_\theta$  denote mean and covariance under the edge probability measure  $w_{uv}(\theta) = e^{\theta^\top z_{uv}} / LO_G(\theta)$ .*

Then  $F_G$  is convex, with

$$\nabla F_G(\theta) = \mathbb{E}_\theta[z], \quad \nabla^2 F_G(\theta) = \text{Cov}_\theta(z) \succeq 0.$$

In particular  $\partial_\gamma F_G = \mathbb{E}_\theta[q] \in [0, 1)$  and  $\partial_\gamma^2 F_G = \text{Var}_\theta(q) \geq 0$ , so  $\log LO$  is nondecreasing and convex in  $\gamma$ , strictly increasing in  $\gamma$  unless every edge of  $G$  joins vertices of equal degree. Moreover  $\text{Var}_\theta(q) = 0$  at some (equivalently, every)  $\theta$  if and only if  $q_{uv}$  is constant over the edges of  $G$ : this covers regular graphs ( $q \equiv 0$ ), but also, e.g.,  $(\Delta, \delta)$ -biregular graphs, on which every edge carries the same nonzero imbalance and  $\log LO$  is affine and strictly increasing in  $\gamma$ .  $F_G$  is strictly convex if and only if the edge feature vectors  $\{z_{uv}\}_{uv \in E}$  do not lie in a proper affine subspace of  $\mathbb{R}^3$ ; on a regular graph  $F_G(\theta) = \log m + \theta^\top z_0$  is affine and  $\text{Cov}_\theta(z) = 0$ .

*Proof.* Direct differentiation of  $LO_G(\theta) = \sum_{uv} e^{\theta^\top z_{uv}}$  gives  $\partial_k F_G = \sum_{uv} w_{uv} z_{uv,k} = \mathbb{E}_\theta[z_k]$  and  $\partial_k \partial_l F_G = \mathbb{E}_\theta[z_k z_l] - \mathbb{E}_\theta[z_k] \mathbb{E}_\theta[z_l]$ , the  $(k, l)$ -entry of  $\text{Cov}_\theta(z)$ , which is positive semidefinite; convexity follows. The measure  $w(\theta)$  has full support on  $E$  for every  $\theta$ , so  $\text{Cov}_\theta(z)$  is singular in a direction  $v$  iff  $v^\top z_{uv}$  is constant over all edges, i.e. iff the  $z_{uv}$  lie in a proper affine subspace; this proves the strictness criterion. Since  $q_{uv} \in [0, 1)$ ,  $\mathbb{E}_\theta[q] \in [0, 1)$ , and  $\mathbb{E}_\theta[q] = 0$  iff  $q_{uv} = 0$  on every edge. On an  $r$ -regular graph all  $z_{uv}$  equal  $z_0 = (2 \ln r, \ln 2r, 0)$ . ■

Theorem 3 places the family in the classical exponential-family framework: moving  $\gamma$  reweights the edge measure toward imbalanced edges at rate  $\mathbb{E}_\theta[q]$ , and the Cauchy–Schwarz midpoint inequality of Section 4 is precisely midpoint log-convexity, i.e. a corollary of this theorem.

### 3.1 Size–shape factorisation, the ratio plane, crossings and identifiability

Since  $4ij/(i+j)^2 = 1 - q_{ij}^2$ , every admissible pair satisfies  $\ln(ij) = 2 \ln(i+j) + \ln(1 - q_{ij}^2) - \ln 4$ , and therefore

$$LO(G; \alpha, \beta, \gamma) = 4^{-\alpha} \sum_{uv \in E(G)} (d_u + d_v)^{2\alpha+\beta} (1 - q_{uv}^2)^\alpha e^{\gamma q_{uv}}. \quad (2)$$

The combination  $\kappa = 2\alpha + \beta$  acts only on the edge degree sum (size), while  $\alpha$  and  $\gamma$  act only on  $q_{uv}$ , which is a function of the degree ratio alone (shape). Write  $n_q(G)$  for the number of edges of  $G$  with  $q_{uv} = q$  (the  $q$ -histogram), and  $\Pi = \{(\alpha, \beta, \gamma) : \beta = -2\alpha\}$  for the plane  $\kappa = 0$ , which contains every pure- $\gamma$  point  $(0, 0, \gamma)$  as well as  $GA = 2LO(\frac{1}{2}, -1, 0)$  and  $AG = \frac{1}{2}LO(-\frac{1}{2}, 1, 0)$ .

**Theorem 4** (Ratio plane). *Let  $G$  and  $H$  be graphs.*

- (i) *On  $\Pi$ ,  $LO(G; \alpha, -2\alpha, \gamma) = \sum_q n_q(G) ((1 - q^2)/4)^\alpha e^{\gamma q}$  depends on  $G$  only through its  $q$ -histogram.  $G$  and  $H$  agree at every point of  $\Pi$  if and only if  $n_q(G) = n_q(H)$  for all  $q$ ; otherwise their values differ at every point of  $\Pi$  outside a set of measure zero.*
- (ii) *For every nonzero algebraic  $\gamma$ ,  $LO(G; 0, 0, \gamma) = LO(H; 0, 0, \gamma)$  if and only if  $n_q(G) = n_q(H)$  for all  $q$ . On any finite set of graphs the same holds for all but finitely many real  $\gamma$ .*
- (iii) *Let  $G$  and  $H$  be trees with distinct edge-degree-pair profiles and equal  $q$ -histograms, each having a leaf adjacent to a vertex of the same degree  $d \geq 2$ . Attaching a pendant path of length  $L \geq 1$  at such a leaf in each produces trees with the same two properties and the same maximum degree. Consequently, for every  $n \geq 13$  (and every  $n \geq 16$  among trees with  $\Delta \leq 4$ ) there are non-isomorphic trees of order  $n$  with distinct degree-pair profiles that no point of  $\Pi$  separates.*

*Proof.* (i) Setting  $\beta = -2\alpha$  in (2) removes the size factor. Agreement for equal histograms is immediate. If the histograms differ, at  $\alpha = 0$  the difference is  $\sum_q (n_q(G) - n_q(H)) e^{\gamma q}$ , an exponential sum with distinct exponents and not all coefficients zero, which is not identically zero; the difference on  $\Pi$  is therefore a nonzero real-analytic function of  $(\alpha, \gamma)$ , whose zero set has measure zero. (ii) For algebraic  $\gamma \neq 0$  the exponents  $\gamma q$ ,  $q \in \mathbb{Q}$ , are distinct algebraic numbers, so by the Lindemann–Weierstrass theorem [6] the numbers  $e^{\gamma q}$  are linearly independent over the algebraic numbers; the integer coefficients  $n_q(G) - n_q(H)$  must all vanish. For real  $\gamma$ , a nonzero exponential sum with  $k$  distinct real exponents has at most  $k - 1$  real zeros (Proposition 5), and a finite set of graphs gives finitely many

such sums. (iii) The attachment changes the profile of each tree by the same vector,  $-(1, d) + (2, d) + (L - 1)(2, 2) + (1, 2)$ , so distinct profiles stay distinct, equal histograms stay equal, and the new or changed vertices have degree at most 2, so the maximum degree is unchanged when it is at least 2. The order-13 pair of Section 5.2 has a common leaf-neighbour degree  $d = 3$ , and the order-16 pair with  $\Delta \leq 4$ ,  $\{(1, 2)^2, (1, 4)^8, (2, 4)^2, (4, 4)^3\}$  and  $\{(1, 4)^8, (2, 2)^3, (2, 4)^4\}$ , has  $d = 4$ ; this gives every larger order. ■

By exhaustive enumeration,  $q$ -histograms coincide with degree-pair profiles for all trees of order at most 12 and for trees with  $\Delta \leq 4$  of order at most 15, so the pure- $\gamma$  points attain the degree-pair floor exactly up to those orders and, by part (iii), at no larger order. Theorem 4 is consistent with the tie of  $GA$ ,  $AG$  and the pure- $\gamma$  points on the trees of order 10 (Section 5.2): all lie on  $\Pi$ , so none can separate more than the  $q$ -histograms (that they separate all of them there is a computational fact). It also shows the limits of the pure- $\gamma$  line: from order 13 generic points off  $\Pi$  (Proposition 15) and the Randić index (which itself leaves the floor only at order 14) separate more trees than any point of  $\Pi$ . In the other direction,  $GA$ , whose values are sums of square roots, merges two distinct  $q$ -histograms among trees first at order 17 (exact witness:  $\{(1, 5)^{10}, (2, 2)^2, (5, 5)^2, (2, 5), (1, 2)\}$  and  $\{(4, 5)^3, (1, 4)^5, (1, 5)^6, (2, 5), (1, 2)\}$  have equal  $GA$  but different  $q$ -histograms), where the pure- $\gamma$  points, by part (ii), still separate all histograms. That  $GA$  depends only on the degree ratio is well known; the content here is the exact ceiling and the colliding families.

Along the  $\gamma$  axis the number of times two graphs can exchange order is bounded by a classical rule of signs.

**Proposition 5** ( $\gamma$ -crossings). *Fix  $(\alpha, \beta)$  and graphs  $G, H$ , and let  $q_1 < \dots < q_r$  be the distinct imbalances of the degree pairs occurring in  $G$  or  $H$ . Write  $D(\gamma) = LO(G; \alpha, \beta, \gamma) - LO(H; \alpha, \beta, \gamma) = \sum_{k=1}^r c_k e^{\gamma q_k}$  with  $c_k = \sum_{q_{ij}=q_k} (m_{ij}(G) - m_{ij}(H))(ij)^\alpha (i+j)^\beta$ , and let  $V$  be the number of sign changes in  $(c_1, \dots, c_r)$  after zero terms are removed. If  $D \not\equiv 0$ , then  $D$  has at most  $V \leq r - 1$  real zeros counted with multiplicity, and their number has the parity of  $V$ . In particular, for graphs with  $\Delta \leq 4$ , where the possible imbalances are  $0, \frac{1}{7}, \frac{1}{5}, \frac{1}{3}, \frac{1}{2}, \frac{3}{5}$ , two graphs exchange order at*

most five times as  $\gamma$  ranges over  $\mathbb{R}$ , for every  $(\alpha, \beta)$ .

*Proof.* With  $x = e^\gamma > 0$ ,  $D$  becomes the generalised polynomial  $\sum_k c_k x^{q_k}$  with distinct real exponents, to which Laguerre's extension of Descartes' rule of signs applies [30]: the number of positive zeros, with multiplicity, is at most  $V$  and has its parity. The parity statement also follows from the signs of  $D$  as  $\gamma \rightarrow \mp\infty$ , which are those of the first and last nonzero  $c_k$  (Proposition 13). ■

The hypothesis  $D \not\equiv 0$  excludes in particular graphs with equal degree-pair profiles, which are tied for every  $\gamma$ . The sign-change bound can be attained: two trees of order 12 with maximum degree 5 and seven distinct imbalances (bound 6) cross five times at  $(\alpha, \beta) = (-1, 0)$ ; for  $\Delta \leq 4$  the largest number of crossings we found is four, so whether five is attained there is open. The proposition concerns the  $\gamma$  axis only: along other directions the exponents are not the  $q_k$  and the constant five does not apply.

Finally, the per-graph identifiability noted in Remark 1 can be stated exactly. Let  $R(G)$  be the set of degree pairs realised by  $G$ ,  $r(G)$  the number of distinct imbalances among them, and  $A(G)$  the affine dimension of  $\{z_{ij} : (i, j) \in R(G)\}$ .

**Proposition 6** (Per-graph scale-invariant directions). *(i)  $A(G)$  equals the rank of the Hessian  $\text{Cov}_\theta(z)$  of  $\log LO_G$  at every  $\theta$ , and the directions  $v$  with  $LO(G; \theta + tv) = e^{ct} LO(G; \theta)$  for all  $t$  and  $\theta$  form a subspace of dimension  $3 - A(G)$ .*

*(ii)  $A(G) \leq \min(|R(G)| - 1, r(G), 3)$  for every graph.*

*(iii) If  $r(G) = 2$  with imbalances  $q_1 \neq q_2$ , then  $v = (1, -2, c)$  with  $c = [\ln(1 - q_2^2) - \ln(1 - q_1^2)]/(q_1 - q_2)$  is such a direction.*

*Proof.* (i) is the standard minimal-representation property of exponential families [9]:  $\text{Cov}_\theta(z)$  has full support on  $R(G)$ , so its rank is the affine dimension of the support, and  $v^\top z$  is constant on  $R(G)$  exactly for  $v$  orthogonal to the differences of the  $z_{ij}$ . (ii) By (2),  $z_{ij} = (2s + \ln(1 - q^2) - \ln 4, s, q)$  with  $s = \ln(i + j)$ , so pairs with the same imbalance lie

on one line parallel to  $(2, 1, 0)$ , and  $r$  parallel lines span an affine space of dimension at most  $r$ . (iii)  $v^\top z_{ij} = \ln(1 - q_{ij}^2) - \ln 4 + c q_{ij}$  takes the same value at  $q_1$  and  $q_2$  by the choice of  $c$ . ■

Equality holds in (ii) for every set of admissible pairs with  $\Delta \leq 6$  (exhaustive check in 60-digit arithmetic), hence for all molecular graphs; it fails for some pair sets from  $\Delta = 7$ , e.g.  $\{(1, 7), (2, 6), (3, 5), (4, 4)\}$ ; every failing set we found has a constant degree sum, which no connected graph realises with four distinct pairs. For example, the spider with four legs of length three realises the pairs  $(2, 4)$ ,  $(2, 2)$  and  $(1, 2)$ , which have only two imbalances ( $\frac{1}{3}$  and 0); on it,  $\gamma$  is confounded with  $(\alpha, \beta)$  along the direction of part (iii). For a QSPR dataset the relevant set is the union of the pairs realised by its molecules, since only a direction along which every molecule is rescaled by the same factor leaves correlations unchanged; on the octanes this union contains nine of the ten admissible pairs and its affine dimension is 3, so no parameter direction is confounded.

## 4 Bounds for the Loyola family

We give a bracket bound via the edge-contribution function  $\psi$ , bounds in  $(m, \Delta, \delta)$  for non-negative parameters, a Cauchy–Schwarz interpolation bound, a uniform ratio bound, and the  $\gamma \rightarrow \pm\infty$  asymptotics; the  $\gamma \neq 0$  bounds rest on bounding  $e^{\gamma q_{uv}}$  edge by edge between 1 and  $e^{\gamma(\Delta-1)/(\Delta+1)}$ . Throughout,  $G$  is a simple connected graph with  $\delta \leq d_v \leq \Delta$  for every  $v \in V(G)$ . Several of the inequalities have been machine-checked (Reproducibility paragraph, Section 7).

*Remark 2* (Attribution at  $\gamma = 0$ ). At  $\gamma = 0$  the family coincides with  $M_{\alpha, \beta}$ , for which inequalities of all the types developed here are already known:  $(\Delta, \delta, m)$ -bounds with regular/biregular equality and two-sided comparisons against  $M_1$ ,  $M_2$ ,  $R$ ,  $\chi$ ,  $H$ ,  $GA$ ,  $AG$  and  $ISI$  [38], Jensen- and Hölder-type relations [19], and further inequalities [42]. The  $\gamma = 0$  specializations below are therefore not claimed as new; the content of this section is their extension to  $\gamma \neq 0$ , where the imbalance factor  $e^{\gamma q_{uv}}$  enters every edge weight, and the tracking of how  $\gamma$  deforms the equality cases

(the  $\gamma = 0$  comparisons with the Randić, sum-connectivity and Sombor indices are not repeated).

The summand in (1) depends only on the unordered edge degree pair  $(d_u, d_v)$ . We isolate this dependence as follows:

**Definition 2.** For real parameters  $\alpha, \beta, \gamma$  and positive reals  $x, y$ ,

$$\psi(x, y; \alpha, \beta, \gamma) = (xy)^\alpha (x + y)^\beta \exp\left(\gamma \frac{|x - y|}{x + y}\right). \quad (3)$$

With this notation  $LO(G; \alpha, \beta, \gamma) = \sum_{uv \in E(G)} \psi(d_u, d_v; \alpha, \beta, \gamma)$ .

**Proposition 7.** *Let  $G$  be a simple connected graph of size  $m \geq 1$  with degrees in  $[\delta, \Delta]$ . Then*

$$m \cdot \psi_{\min}(\delta, \Delta; \alpha, \beta, \gamma) \leq LO(G; \alpha, \beta, \gamma) \leq m \cdot \psi_{\max}(\delta, \Delta; \alpha, \beta, \gamma), \quad (4)$$

where,  $\psi_{\min} = \min\{\psi(i, j; \alpha, \beta, \gamma) : \delta \leq i \leq j \leq \Delta\}$  and  $\psi_{\max} = \max\{\psi(i, j; \alpha, \beta, \gamma) : \delta \leq i \leq j \leq \Delta\}$ . The upper inequality is an equality if and only if every edge of  $G$  has a degree pair attaining  $\psi_{\max}$ , and the lower inequality is an equality if and only if every edge has a degree pair attaining  $\psi_{\min}$ .

*Proof.* Every edge  $uv$  has its degree pair in the admissible set  $\mathcal{P}(\delta, \Delta) := \{(i, j) \in \mathbb{Z}^2 : \delta \leq i \leq j \leq \Delta\}$ , so  $\psi_{\min} \leq \psi(d_u, d_v; \alpha, \beta, \gamma) \leq \psi_{\max}$  on every edge; summing over the  $m$  edges and using  $\sum_{uv} \psi(d_u, d_v; \alpha, \beta, \gamma) = LO(G; \alpha, \beta, \gamma)$  gives (4). Each side of (4) is the sum of  $m$  per-edge inequalities, so it is an equality if and only if every per-edge inequality is, i.e. if and only if every edge has a degree pair attaining  $\psi_{\max}$  (upper side), respectively  $\psi_{\min}$  (lower side), over  $\mathcal{P}(\delta, \Delta)$ . ■

**Corollary 8.** *Let  $\alpha_1 + \alpha_2 = 2\alpha$ ,  $\beta_1 + \beta_2 = 2\beta$ , and  $\gamma_1 + \gamma_2 = 2\gamma$ . Then for every simple connected graph  $G$  without isolated vertices,*

$$LO(G; \alpha, \beta, \gamma) \leq \sqrt{LO(G; \alpha_1, \beta_1, \gamma_1) \cdot LO(G; \alpha_2, \beta_2, \gamma_2)}$$

*Equality holds if and only if the ratio*

$$\psi(d_u, d_v; \alpha_1, \beta_1, \gamma_1) / \psi(d_u, d_v; \alpha_2, \beta_2, \gamma_2)$$

is the same for every edge of  $G$ .

*Proof.* For each edge  $e = uv$  write  $\psi_e^{(k)} := \psi(d_u, d_v; \alpha_k, \beta_k, \gamma_k) > 0$ ,  $k \in \{1, 2\}$ . By additivity of the exponents and the hypotheses on  $(\alpha_k, \beta_k, \gamma_k)$ ,  $\psi_e^{(1)}\psi_e^{(2)} = \psi(d_u, d_v; \alpha, \beta, \gamma)^2$  on every edge. The Cauchy–Schwarz inequality for  $a_e = (\psi_e^{(1)})^{1/2}$  and  $b_e = (\psi_e^{(2)})^{1/2}$  gives  $\sum_e a_e b_e \leq (\sum_e a_e^2)^{1/2} (\sum_e b_e^2)^{1/2}$ , which is the claim because  $\sum_e a_e b_e = LO(G; \alpha, \beta, \gamma)$ ,  $\sum_e a_e^2 = LO(G; \alpha_1, \beta_1, \gamma_1)$  and  $\sum_e b_e^2 = LO(G; \alpha_2, \beta_2, \gamma_2)$ . Equality holds if and only if  $b_e = \lambda a_e$  for a constant  $\lambda \geq 0$  and all  $e$ , i.e. if and only if  $\psi_e^{(2)}/\psi_e^{(1)} = \lambda^2$  is the same on every edge. (The inequality itself is midpoint log-convexity of  $\theta \mapsto LO(G; \theta)$ , Theorem 3.) ■

**Proposition 9.** *Let  $G$  be a connected graph of size  $m$  and maximum degree  $\Delta$ . For  $\alpha, \beta \geq 0$  and  $\gamma \geq 0$ ,*

$$LO(G; \alpha, \beta, \gamma) \leq m \cdot \Delta^{2\alpha} \cdot (2\Delta)^\beta \cdot \exp\left(\gamma \frac{\Delta - 1}{\Delta + 1}\right). \quad (5)$$

*Proof.* On every edge  $uv$  we have  $1 \leq d_u, d_v \leq \Delta$ , hence  $d_u d_v \leq \Delta^2$  and  $d_u + d_v \leq 2\Delta$ ; and the normalised imbalance  $\text{imb}(x, y) = |x - y|/(x + y)$  is maximised over  $\{(i, j) : 1 \leq i \leq j \leq \Delta\}$  at  $(1, \Delta)$ , so  $|d_u - d_v|/(d_u + d_v) \leq (\Delta - 1)/(\Delta + 1)$ . Since  $\alpha, \beta, \gamma \geq 0$ , the maps  $t \mapsto t^\alpha$ ,  $t \mapsto t^\beta$  and  $t \mapsto e^{\gamma t}$  are non-decreasing, so  $\psi(d_u, d_v; \alpha, \beta, \gamma) \leq \Delta^{2\alpha} (2\Delta)^\beta \exp(\gamma(\Delta - 1)/(\Delta + 1))$ ; summing over the  $m$  edges yields (5). For  $\gamma > 0$  and  $\Delta \geq 2$  the bound is attained if and only if  $\alpha = \beta = 0$  and every edge has degree pair  $(1, \Delta)$  (for connected  $G$ ,  $G \cong K_{1, \Delta}$ ); if  $\alpha > 0$  or  $\beta > 0$  the three factor-wise maxima are not simultaneously attainable and the inequality is strict. ■

**Proposition 10.** *Let  $G$  be a connected graph of size  $m$  and minimum degree  $\delta$ . For  $\alpha, \beta \geq 0$  and  $\gamma \geq 0$ ,*

$$LO(G; \alpha, \beta, \gamma) \geq m \cdot \delta^{2\alpha} \cdot (2\delta)^\beta, \quad (6)$$

*with equality if and only if  $G$  is  $\delta$ -regular, except in the degenerate case  $(\alpha, \beta, \gamma) = (0, 0, 0)$ , where  $LO(G; \alpha, \beta, \gamma) \equiv m$  and the bound holds with equality for every graph.*

*Proof.* On every edge  $d_u, d_v \geq \delta$ , so for  $\alpha, \beta \geq 0$ ,  $(d_u d_v)^\alpha \geq \delta^{2\alpha}$  and  $(d_u + d_v)^\beta \geq (2\delta)^\beta$ , and for  $\gamma \geq 0$  the exponential factor is at least 1 because the imbalance is non-negative. Multiplying and summing over the  $m$  edges yields (6). If  $(\alpha, \beta, \gamma) = (0, 0, 0)$  then  $LO(G; 0, 0, 0) = m$  equals the bound for every graph. Otherwise, equality requires equality in every active per-edge factor: if  $\alpha > 0$  or  $\beta > 0$  this forces  $d_u = d_v = \delta$  on every edge; if  $\alpha = \beta = 0$  and  $\gamma > 0$  it forces  $d_u = d_v$  on every edge, so the connected graph  $G$  is regular, hence  $\delta$ -regular. Conversely, on a  $\delta$ -regular graph every edge has degree pair  $(\delta, \delta)$  and equality holds. ■

The next theorem gives a single uniform template for bounding  $LO(G; \alpha, \beta, \gamma)$  both above and below in terms of any classical bond-incident-degree index whose edge contribution is strictly positive on the admissible degree pairs of the graph.

**Proposition 11.** *Let  $G$  be a simple graph with maximum degree at most  $\Delta$ , and let  $I_h(G) = \sum_{uv \in E(G)} h(d_u, d_v)$  be a BID index with edge contribution  $h(i, j)$  strictly positive on every degree pair  $(i, j)$  that appears in  $G$ . Define*

$$c_{\min}^h = \min_{(i,j)} \frac{\psi(i, j; \alpha, \beta, \gamma)}{h(i, j)}, \quad c_{\max}^h = \max_{(i,j)} \frac{\psi(i, j; \alpha, \beta, \gamma)}{h(i, j)},$$

where the extrema range over the degree pairs actually appearing in  $G$ . Then

$$c_{\min}^h \cdot I_h(G) \leq LO(G; \alpha, \beta, \gamma) \leq c_{\max}^h \cdot I_h(G), \quad (7)$$

with equality on either side if and only if every edge of  $G$  attains the corresponding extremum of  $\psi/h$ .

*Proof.* Since  $h(d_u, d_v) > 0$  and  $c_{\min}^h \leq \psi(d_u, d_v; \alpha, \beta, \gamma)/h(d_u, d_v) \leq c_{\max}^h$  on every edge  $uv$ , multiplying by  $h(d_u, d_v)$  and summing over  $E(G)$ , with  $\sum_{uv} h(d_u, d_v) = I_h(G)$  and  $\sum_{uv} \psi = LO(G; \alpha, \beta, \gamma)$ , gives (7); each side is an equality if and only if every edge attains the corresponding extremum of  $\psi/h$  over the appearing pairs. ■

Proposition 11 is a *sharp graph-specific* comparison: its constants depend on the degree pairs that actually appear in  $G$ . A uniform version

follows immediately by enlarging the range of the extrema.

**Corollary 12** (Uniform  $\Delta$ -bounded version). *With the hypotheses of Proposition 11, but with  $h$  strictly positive on all of  $P_\Delta = \{(i, j) : 1 \leq i \leq j \leq \Delta\}$  and the extrema taken over  $P_\Delta$ , inequality (7) holds simultaneously for every graph of maximum degree at most  $\Delta$ , with constants independent of the individual graph. Equality on either side holds if and only if every edge of  $G$  realises a degree pair attaining the corresponding extremum over  $P_\Delta$ ; the uniform constants are in general strictly worse than the graph-specific ones, since the extrema are taken over a superset.*

**Proposition 13.** *At  $\gamma = 0$ ,  $LO(G; \alpha, \beta, 0)$  coincides with the two-parameter  $(\alpha, \beta)$  degree-based index  $\sum_{uv \in E(G)} (d_u d_v)^\alpha (d_u + d_v)^\beta$ . Let  $E_{\max}(G)$  denote the set of edges realising  $\text{imb}_{\max}(G) = \max_{uv} \text{imb}(d_u, d_v)$  and  $E_{\min}(G)$  the set realising  $\text{imb}_{\min}(G) = \min_{uv} \text{imb}(d_u, d_v)$ . Then as  $\gamma \rightarrow +\infty$ ,*

$$\frac{LO(G; \alpha, \beta, \gamma)}{e^{\gamma \text{imb}_{\max}(G)}} \longrightarrow \sum_{uv \in E_{\max}(G)} (d_u d_v)^\alpha (d_u + d_v)^\beta,$$

and as  $\gamma \rightarrow -\infty$ ,

$$LO(G; \alpha, \beta, \gamma) / e^{\gamma \text{imb}_{\min}(G)} \longrightarrow \sum_{uv \in E_{\min}(G)} (d_u d_v)^\alpha (d_u + d_v)^\beta.$$

*Proof.* At  $\gamma = 0$  the exponential factor is 1 on every edge. With  $\mu(G) := \text{imb}_{\max}(G)$  and  $\varepsilon_{uv} := \mu(G) - \text{imb}(d_u, d_v) \geq 0$ ,  $LO(G; \alpha, \beta, \gamma) / e^{\gamma \mu(G)} = \sum_{uv} (d_u d_v)^\alpha (d_u + d_v)^\beta e^{-\gamma \varepsilon_{uv}}$ ; the summands over  $E_{\max}(G)$  ( $\varepsilon_{uv} = 0$ ) do not depend on  $\gamma$ , and the others ( $\varepsilon_{uv} > 0$ ) tend to 0 as  $\gamma \rightarrow +\infty$ , termwise since  $E(G)$  is finite. The case  $\gamma \rightarrow -\infty$  is dual. ■

## 5 Discrimination and the degree-pair information ceiling

### 5.1 The degree-pair ceiling

**Proposition 14** (Degree-pair information ceiling). *If two graphs  $G$  and  $H$  have identical unordered edge-degree-pair count vectors,  $m_{ij}(G) = m_{ij}(H)$  for all  $i \leq j$ , then  $LO(G; \theta) = LO(H; \theta)$  for every  $\theta \in \mathbb{R}^3$ , and more generally  $TI(G) = TI(H)$  for every vertex-degree-based index  $TI$ . Among the 18 octane isomers exactly 16 distinct count vectors occur: the pairs  $\{3\text{-methylheptane}, 4\text{-methylheptane}\}$  and  $\{3,4\text{-dimethylhexane}, 3\text{-ethyl-2-methylpentane}\}$  share their count vectors while differing in measured properties (Table 1), so every vertex-degree-based descriptor, tuned or not, carries a nonzero irreducible error on this benchmark.*

*Proof.* By Definition 1,  $LO(G; \theta)$  depends on  $G$  only through the multiset of unordered edge degree pairs, i.e. through  $(m_{ij}(G))_{i \leq j}$ ; the same holds for every vertex-degree-based index [44]. The counts for the 18 octane isomers are computed and cross-checked in the reproducibility package; both collision pairs have their common count vectors listed in Table 1. ■

The ceiling is tight in the following sense: distinct count vectors are separated by the family at almost every parameter choice.

**Proposition 15** (Generic discrimination of distinct profiles). *Fix  $\Delta \geq 1$  and let  $m \neq m'$  be two distinct edge-degree-pair count vectors supported on the admissible pairs  $P_\Delta$ . Then  $D(\theta) = \sum_{(i,j) \in P_\Delta} (m_{ij} - m'_{ij}) e^{\theta^\top z_{ij}}$  is a real-analytic function of  $\theta$  that is not identically zero, and its zero set has Lebesgue measure zero in  $\mathbb{R}^3$ . Consequently, any two graphs with distinct count vectors are distinguished by  $LO(\cdot; \theta)$  for almost every  $\theta \in \mathbb{R}^3$ ; only graphs with identical count vectors (Proposition 14) are indistinguishable for all  $\theta$ .*

*Proof.* The map  $(i, j) \mapsto z_{ij}$  is injective on  $P_\Delta$ , since  $\ln(ij)$  and  $\ln(i + j)$  recover the product and the sum, which determine the unordered pair. Choose  $v \in \mathbb{R}^3$  such that the inner products  $v^\top z_{ij}$  are pairwise distinct (this excludes finitely many hyperplanes of  $v$ 's). Along the ray

$\theta = tv$ ,  $D(tv) = \sum_k c_k e^{t\lambda_k}$  with distinct exponents  $\lambda_k$  and coefficients  $c_k = m_{i_k j_k} - m'_{i_k j_k}$ , not all zero; such an exponential sum vanishes identically only if all  $c_k = 0$ , so  $D \not\equiv 0$  on the ray and hence on  $\mathbb{R}^3$ .  $D$  is a finite sum of exponentials of linear functionals, hence real-analytic on  $\mathbb{R}^3$ , and the zero set of a nonzero real-analytic function has Lebesgue measure zero. ■

The proof uses only the injectivity of  $(i, j) \mapsto (\ln(ij), \ln(i + j))$ , so the same conclusion already holds for the parent family  $M_{\alpha, \beta}$  on the  $(\alpha, \beta)$ -plane. Proposition 15 is therefore a property the Loyola family inherits, not an advantage conferred by  $\gamma$ . Likewise, since  $q_{ij} = \sqrt{1 - 4ij/(i + j)^2}$  is a function of the pair, the imbalance coordinate adds no graph information beyond the edge-degree-pair counts (every vertex-degree-based index is a functional of those counts [44]); what  $\gamma$  adds is the new functional direction of Proposition 2, i.e. a different weighting of the same information.

**Table 1.** The two degree-pair collision pairs among the 18 octane isomers. Each pair shares its complete edge-degree-pair count vector (shown as multiplicities of the unordered pairs), hence is indistinguishable by every vertex-degree-based index for all parameter values, yet differs in the measured properties.

| Isomer                                                         | $T_B$ | $\Delta H_f$ | $\Delta H_{\text{vap}}$ | $S$    | $\omega$ |
|----------------------------------------------------------------|-------|--------------|-------------------------|--------|----------|
| <i>shared count vector</i> 2(1, 2), 1(1, 3), 2(2, 2), 2(2, 3): |       |              |                         |        |          |
| 3-methylheptane                                                | 118.9 | -50.82       | 9.52                    | 111.26 | 0.371    |
| 4-methylheptane                                                | 117.7 | -50.69       | 9.48                    | 109.32 | 0.372    |
| <i>shared count vector</i> 2(1, 2), 2(1, 3), 2(2, 3), 1(3, 3): |       |              |                         |        |          |
| 3,4-dimethylhexane                                             | 117.7 | -50.91       | 9.32                    | 106.59 | 0.340    |
| 3-ethyl-2-methylpentane                                        | 115.6 | -50.48       | 9.21                    | 106.06 | 0.330    |

## 5.2 First loss of discrimination across tree orders

Degeneracy measures an index’s ability to discriminate non-isomorphic graphs [32]; lower is better. On the 106 non-isomorphic trees of order 10, with the distinct-value convention  $100(1 - k/N)$  ( $k$  distinct values among  $N = 106$  trees; Supplementary Section S5), the pure- $\gamma$  points  $LO(0, 0, 1)$  and  $LO(0, 0, 2)$  attain degeneracy 25.5% (79 distinct values), tied with  $R$ ,  $GA$  and  $AG$ ; since the 106 trees realise only 79 distinct

edge-degree-pair count vectors, 25.5% is the minimum attainable by *any* degree-pair-based index on this set, while  $M_1$  (83.0%) and  $M_2$  (67.0%) are the most degenerate (Supplementary Figure S2). Repeating the computation for all trees of orders 7 through 12 and for their molecular subsets with  $\Delta \leq 4$  (Supplementary Table S4), the pure- $\gamma$  points attain the degree-pair-profile floor at every order in both sets; so does the Randić index (while  $\chi$  falls off the floor from order 10), so floor attainment distinguishes the pure- $\gamma$  points from  $M_1$  and  $M_2$  but not from  $R$ . Theorem 4 turns this into an exact statement: the pure- $\gamma$  points separate precisely the  $q$ -histograms, which coincide with the degree-pair profiles up to order 12 (order 15 for  $\Delta \leq 4$ ) and not beyond. Among the trees of order 13 the profiles  $\{(1, 2)^1, (1, 3)^2, (1, 6)^5, (2, 3)^2, (2, 6)^1, (3, 3)^1\}$  and  $\{(1, 3)^3, (1, 6)^5, (2, 2)^1, (2, 3)^2, (3, 6)^1\}$  (exponents are edge multiplicities) share the  $q$ -histogram  $\{0^1, \frac{1}{5}^2, \frac{1}{3}^1, \frac{1}{2}^3, \frac{5}{7}^5\}$ . Table 2 collects the first failures; at order 14, where  $R$  falls below the floor, the pure- $\gamma$  points give 1090 values, and  $R$  still attains the floor at order 13 (570 values).

**Table 2.** First tree order at which discrimination is lost. “Below floor”: first order with fewer distinct values than distinct edge-degree-pair profiles (the floor for every vertex-degree-based index), with (values/profiles) at that order. “Merges”: first order at which two trees with different  $q$ -histograms get equal values.  $LO(0, 0, \gamma)$ : the pure- $\gamma$  points  $\gamma = 1, 2$ . Orders below 7 not examined; “—”: not determined.

| Index              | all trees                 |                    | trees with $\Delta \leq 4$ |                    |
|--------------------|---------------------------|--------------------|----------------------------|--------------------|
|                    | below floor               | merges             | below floor                | merges             |
| $LO(0, 0, \gamma)$ | 13 (566/570)              | never <sup>a</sup> | 16 (809/810)               | never <sup>a</sup> |
| $R$                | 14 (1095/1100)            | n/a <sup>b</sup>   | > 12 <sup>c</sup>          | n/a <sup>b</sup>   |
| $GA$               | 13 (566/570) <sup>d</sup> | 17 <sup>e</sup>    | 16 (809/810) <sup>d</sup>  | — <sup>e</sup>     |

<sup>a</sup> All  $q$ -histograms are separated for every nonzero algebraic  $\gamma$  (Theorem 4(ii)). <sup>b</sup>  $R$  is not on  $\Pi$ . <sup>c</sup>  $R$  attains the floor for  $n \leq 12$  in both sets; larger molecular trees were not tabulated for  $R$ . <sup>d</sup>  $GA$  lies on  $\Pi$  and merges no  $q$ -histograms below order 17, so there it ties the pure- $\gamma$  points. <sup>e</sup> The witness (Section 3.1) has maximum degree 5.

### 5.3 Structure sensitivity on decane isomers

The structure-sensitivity criterion of Furtula, Gutman and Dehmer [17] (with a fingerprint-based variant due to Rakić and Furtula [45]) was implemented with the published protocol on the 75 decane isomer trees, the similar structures of a tree being those obtained by relocating a single edge within the molecular class; *SS* and *Abr* are the class means of the mean and the maximum relative change of the index over these neighbours. The implementation reproduces the published decane-class values of Barman and Das [7] (eleven of fourteen *SS/Abr* values exactly at the four displayed decimals, the other three within one unit in the fourth); the protocol and this control are detailed in Supplementary Section S6. Under the published desideratum (*SS* large, *Abr* small), *LO*(0,0,2) has a structure sensitivity equal to that of the hyper-Zagreb index (0.1257 against 0.1256) at a markedly smaller abruptness (0.274 against 0.313), and the sensitivity-to-abruptness ratios of the pure- $\gamma$  points (0.4579 and 0.4565) are comparable to those of *H/2*, *GA* and *R* (0.448–0.451; Table 3); in view of the redundancy noted in Section 6.1, these third-decimal differences are not a ranking. Moreover, *SS* is not a fixed property of the family: along the pure- $\gamma$  line it grows with  $\gamma$  (0.031, 0.062, 0.126, 0.319 at  $\gamma = 0.5, 1, 2, 5$ ) while the ratio peaks near  $\gamma = 2$  and then falls (0.427 at  $\gamma = 5$ ); and the parameter-free aggregate *IRLA*, which the pure- $\gamma$  points approximate to first order in  $\gamma$ , has a higher ratio (*SS* = 0.2343, *Abr* = 0.5066, ratio 0.4625). The favourable sensitivity–abruptness balance is therefore a property of linear imbalance weighting rather than of the tuned exponential, and the *SS* margin over hyper-Zagreb is a tie, not a ranking.

## 6 A concise QSPR diagnostic

We benchmark the Loyola family against the classical indices it reduces to on the 18 octane isomers, following the chemometric benchmark of the diminished Sombor study [40], and on the 35 nonane isomers as an external set. The comparison set is restricted to the exact reductions of Section 1 —

**Table 3.** Structure sensitivity  $SS$ , abruptness  $Abr$ , and their ratio on the 75 decane isomers under the published similar-structure protocol of [17] (single-edge relocation within the molecular class).

| Index         | $SS$   | $Abr$  | $SS/Abr$ |
|---------------|--------|--------|----------|
| $M_1$         | 0.0566 | 0.1336 | 0.4238   |
| $M_2$         | 0.0922 | 0.2443 | 0.3776   |
| $HM$          | 0.1256 | 0.3126 | 0.4017   |
| ${}^mM_2$     | 0.0516 | 0.1177 | 0.4384   |
| $R$           | 0.0264 | 0.0588 | 0.4481   |
| $\chi$        | 0.0264 | 0.0591 | 0.4463   |
| $H/2$         | 0.0516 | 0.1143 | 0.4514   |
| $ISI$         | 0.0226 | 0.0597 | 0.3795   |
| $GA$          | 0.0241 | 0.0535 | 0.4507   |
| $AG$          | 0.0256 | 0.0579 | 0.4415   |
| $LO(0, 0, 1)$ | 0.0624 | 0.1367 | 0.4565   |
| $LO(0, 0, 2)$ | 0.1257 | 0.2745 | 0.4579   |

$M_1, M_2, HM, {}^mM_2, R, \chi, H/2, ISI, GA, AG$  — together with the two fixed pure- $\gamma$  benchmark points  $LO(0, 0, 1)$  and  $LO(0, 0, 2)$  chosen for this study (the edge count  $m = LO(0, 0, 0)$  is constant on the octanes and omitted); no parameter in the correlation comparison is tuned against the property data (an exploratory local-tuning check, Supplementary Section S2, did not exceed the best fixed benchmark except by a 0.001 in-sample  $\Delta H_{\text{vap}}$  margin). All comparisons are descriptive and diagnostic: on these small sets they show what imbalance information does and does not add to the parent family, not chemical predictive utility, and given the number of properties, indices, and tuning candidates examined, no confirmatory inference is claimed. Values are computed in double precision and rounded *once* for display (three decimals for correlations,  $Q^2$  and RMSE, four for structure sensitivity), so a tabulated difference may differ by one unit in the last place from the difference of two displayed entries.

## 6.1 Property correlations on the octane isomers

Table 4 reports the Pearson correlation  $r$  between each index and each of the five physicochemical properties of the 18 octane isomers (boiling point  $T_B$ , enthalpy of formation  $\Delta H_f$ , enthalpy of vaporization  $\Delta H_{\text{vap}}$ ,

entropy  $S$ , acentric factor  $\omega$ ; data and units in Appendix A). Property values are traced to the NIST WebBook [35], except the acentric factors (KDB-derived compilations; NIST does not list them); all 90 molecule-level values were audited against authoritative sources (Supplementary Section S1, with the molecule-by-molecule discussion). Three caveats remain. The 298 K vaporization enthalpy of 2,2,3,3-tetramethylbutane, a solid at 298 K (triple point 374 K), is the benchmark-compilation value 8.41 kcal mol<sup>-1</sup> (35.19 kJ mol<sup>-1</sup>), matching neither NIST standard vaporization enthalpy (42.94 and 42.91 kJ mol<sup>-1</sup>, about 10.26 kcal mol<sup>-1</sup>), so its physical reference state is unresolved. Four entropies could not be matched to a primary source; substituting the values of a widely used compilation changes all twelve  $S$  correlations (by up to 0.034 in  $|r|$ ) but not the winning descriptor ( $HM$ ). Alternative compiled acentric factors change several  $\omega$  correlations at the third decimal (e.g.  $M_2$  from  $-0.988$  to  $-0.986$ ) but not the property-wise winning descriptor.

**Table 4.** Pearson correlation  $r$  between the Loyola reductions, the fixed pure- $\gamma$  benchmark points and the five physicochemical properties of the 18 octane isomers. Bold flags the best  $|r|$  per column.

| Index                             | $T_B$         | $\Delta H_f$  | $\Delta H_{\text{vap}}$ | $S$           | $\omega$      |
|-----------------------------------|---------------|---------------|-------------------------|---------------|---------------|
| $M_1 = LO(0, 1, 0)$               | -0.718        | -0.762        | -0.935                  | -0.973        | -0.970        |
| $M_2 = LO(1, 0, 0)$               | -0.497        | -0.542        | -0.811                  | -0.940        | <b>-0.988</b> |
| $HM = LO(0, 2, 0)$                | -0.654        | -0.710        | -0.903                  | <b>-0.975</b> | -0.982        |
| ${}^mM_2 = LO(-1, 0, 0)$          | <b>+0.855</b> | <b>+0.891</b> | +0.925                  | +0.863        | +0.794        |
| $R = LO(-1/2, 0, 0)$              | +0.819        | +0.850        | +0.958                  | +0.933        | +0.898        |
| $\chi = LO(0, -1/2, 0)$           | +0.800        | +0.832        | +0.961                  | +0.949        | +0.924        |
| $H/2 = LO(0, -1, 0)$              | +0.821        | +0.849        | +0.961                  | +0.933        | +0.900        |
| $ISI = LO(1, -1, 0)$              | -0.005        | -0.000        | -0.383                  | -0.601        | -0.738        |
| $GA = 2 LO(1/2, -1, 0)$           | +0.822        | +0.858        | +0.965                  | +0.941        | +0.906        |
| $AG = \frac{1}{2} LO(-1/2, 1, 0)$ | -0.816        | -0.859        | -0.957                  | -0.939        | -0.900        |
| $LO(0, 0, 1)$                     | -0.828        | -0.832        | <b>-0.984</b>           | -0.931        | -0.929        |
| $LO(0, 0, 2)$                     | -0.829        | -0.846        | -0.981                  | -0.939        | -0.925        |

The best correlation per property is split among the reductions and the pure- $\gamma$  points:  ${}^mM_2 = LO(-1, 0, 0)$  is strongest on  $T_B$  and  $\Delta H_f$ ,  $HM = LO(0, 2, 0)$  on  $S$ ,  $M_2 = LO(1, 0, 0)$  on  $\omega$ , and the fixed (not tuned) pure- $\gamma$  benchmark point  $LO(0, 0, 1)$  on  $\Delta H_{\text{vap}}$  ( $|r| = 0.984$ , with

$LO(0, 0, 2)$  second at 0.981). At the exact-reduction triples the Loyola correlations coincide with the corresponding classical indices to floating-point precision, as they must. For each property a descriptive post-selection paired bootstrap interval for the observed top-two  $|r|$  difference included zero (not multiplicity-adjusted; such intervals do not establish equality or equivalence), and recomputing each bolded correlation with single isomers deleted shifts  $|r|$  by at most 0.027; bootstrap details are in Supplementary Section S1.

On the octanes the pure- $\gamma$  points are highly collinear with the classical reductions (maximum  $|r|$  with a reduction equal to 0.982 for  $LO(0, 0, 1)$  and 0.993 for  $LO(0, 0, 2)$ ).

**Descriptor redundancy.** Fourteen of the descriptor pairs have  $|r| \geq 0.99$  and the first two principal components carry 99.05% of the variance (Supplementary Section S1), so differences in the third decimal of  $|r|$  between near-collinear descriptors should not be read as meaningful rankings.

**The information ceiling realised on this benchmark.** Proposition 14 is not hypothetical on these data: the 18 isomers realise only 16 distinct edge-degree-pair count vectors, and the two collision pairs of Table 1 are indistinguishable by *every* vertex-degree-based index at *every* parameter choice, while their measured properties differ (by up to  $1.2^\circ\text{C}$  in  $T_B$  and 1.94 units in  $S$  within the first pair, and  $2.1^\circ\text{C}$  in  $T_B$  within the second). Every correlation in Table 4, and every tuned variant in the sections below, operates above this irreducible floor.

## 6.2 Freeing $\gamma$ gives no consistent gain once the parent is not truncated

Given identical search budgets, does freeing  $\gamma$  improve out-of-sample prediction beyond an optimized parent  $M_{\alpha,\beta}$ ? In a fully nested leave-one-out design (details in Supplementary Section S3), for each property and each outer fold, candidate parameter vectors drawn uniformly per coordinate from a search box (triples  $(\alpha, \beta, \gamma)$  for the Loyola model, pairs  $(\alpha, \beta)$  with  $\gamma = 0$  for the parent; identical budget) are scored on the training isomers

only by *inner* leave-one-out root-mean-square error of the one-descriptor linear regression; the best candidate is refit and the held-out isomer predicted, so no quantity computed on the held-out isomer enters selection or fitting. In the box  $(-2, 2)$  used first, freeing  $\gamma$  appeared to help the vaporization enthalpy (Supplementary Table S3): over one hundred candidate streams it improved nested predictive performance for  $\Delta H_{\text{vap}}$  in every seed (100/100 at budgets 200 and 500; median  $\Delta Q^2 = +0.069$  and  $+0.068$ ), while it usually degraded  $\Delta H_f$  and gave mixed or slightly negative results for the other properties (Supplementary Section S3 also reports the analysis with 2,2,3,3-tetramethylbutane excluded). These counts describe optimizer variance on one fixed set of 18 molecules and carry no inferential weight, and they hide a boundary effect: for  $\Delta H_{\text{vap}}$  the parent chose  $(\alpha, \beta) = (1.021, -1.993)$ , on the boundary of the box, in 17 of the 18 folds.

These comparisons, however, share one design choice that turns out to be decisive: the box  $(-2, 2)$ . Because  $\ln(ij) = 2\ln(i+j) + \ln(1-q^2) - \ln 4$  (equation (2)), the parent family reaches imbalance information through  $(1-q^2)^\alpha$  along the direction  $\beta \approx -2\alpha$ , but it needs large exponents to weight it strongly, and the box truncates exactly that direction. To test this, the matched comparison was repeated (this analysis was not pre-registered) with 2000 candidates per model, seeds 0–49, and boxes of half-width 2 and 12 for both models (Table 5). In the box of half-width 12 the parent’s selections are interior in more than 90% of folds for every property, and the vaporization-enthalpy gain shrinks to about zero: a hybrid control that widens only  $(\alpha, \beta)$  gives median  $\Delta Q^2 = +0.004$  on the octanes (28 of 50 seeds) and  $-0.002$  on the nonanes (24 of 50), and widening all three coordinates gives  $-0.011$  and  $-0.016$ . On the other properties no advantage is consistent across boxes: under the hybrid control the Loyola model is ahead for octane  $S$  (38 of 50 seeds) and nonane  $T_B$  (31 of 50), but behind for octane  $T_B$ ,  $\Delta H_f$  and  $\omega$ , and the wide box itself lowers the parent’s  $Q^2$  for  $S$  and  $T_B$ . The vaporization-enthalpy gains reported above therefore reflect the truncated box rather than information that the parent lacks; what the third coordinate offers is economy, since an exponent near 1 approaches what the parent reaches only near  $(\alpha, \beta) \approx (4.5, -9)$

( $Q^2 = 0.949$  there against 0.957 for  $LO(0, 0, 1)$  on the octanes).

**Table 5.** Sensitivity of the matched comparison to the search box (octanes: oct.,  $n = 18$ ; nonanes: non.). Edge: share of outer folds in which the parent’s selection lies within 10% of the boundary of the box of the stated half-width. Medians of  $Q^2$  over seeds 0–49 at 2000 candidates per model, and the number of seeds (of 50) in which LO beats GM (half-width 2) or  $LO_h$  beats GM (half-width 12).  $LO_h$  is a hybrid control with  $(\alpha, \beta)$  in the wide box and  $\gamma$  in  $(-2, 2)$ , which avoids diluting the Loyola search over a wide  $\gamma$  range.

| Set, prop.                   | half-width 2 |       |       |    | half-width 12 |       |       |        |    |
|------------------------------|--------------|-------|-------|----|---------------|-------|-------|--------|----|
|                              | Edge         | GM    | LO    | >  | Edge          | GM    | LO    | $LO_h$ | >  |
| oct. $T_B$                   | 69%          | 0.665 | 0.730 | 36 | 2%            | 0.606 | 0.495 | 0.543  | 21 |
| oct. $\Delta H_f$            | 52%          | 0.821 | 0.769 | 11 | 0%            | 0.820 | 0.670 | 0.761  | 10 |
| oct. $\Delta H_{\text{vap}}$ | 97%          | 0.887 | 0.957 | 50 | 3%            | 0.942 | 0.934 | 0.946  | 28 |
| oct. $S$                     | 85%          | 0.919 | 0.915 | 20 | 4%            | 0.858 | 0.848 | 0.882  | 38 |
| oct. $\omega$                | 0%           | 0.979 | 0.974 | 10 | 0%            | 0.979 | 0.938 | 0.951  | 7  |
| non. $T_B$                   | 61%          | 0.576 | 0.601 | 30 | 0%            | 0.524 | 0.392 | 0.577  | 31 |
| non. $\Delta H_{\text{vap}}$ | 93%          | 0.855 | 0.895 | 49 | 10%           | 0.892 | 0.881 | 0.889  | 24 |

The honest summary is therefore that tuning  $\gamma$  yields no consistent predictive advantage over a parent model that is allowed to reach its own optimum. Its value lies in the structural results of Section 3 and in the parsimony documented in Section 6.3.

### 6.3 External test on the nonane isomers and stronger baselines

Under a protocol committed to the reproducibility package before any nonane value was collected, the identical nested design was applied to the 35 constitutional nonane isomers, with boiling points (35 NIST averages of experimental determinations) and standard vaporization enthalpies (34 values) from the NIST WebBook [35]; only 5 of the 34 vaporization enthalpies are calorimetric determinations and 29 come from two compilations [34, 54], so agreement with them measures consistency with the compiled values rather than independent experimental validation; within-fold baselines are in Table 6. The pre-registered verdicts are that the  $\gamma$  effect *replicates* for  $\Delta H_{\text{vap}}$  (compilation values; LO better than GM

in 95/100 and 98/100 seeds, median  $\Delta Q^2 = +0.048$  and  $+0.053$ ) and is *mixed* for  $T_B$  (58/100 and 63/100). The pre-registered protocol, however, fixed the box of half-width 2, which binds the parent (Table 5); in a box that does not, the nonane gain disappears (median  $\Delta Q^2 = -0.002$  under the hybrid control), and the pre-registered verdict is reported as obtained, with this deviation recorded. The fixed point  $LO(0, 0, 1)$ , which involves no tuning at all, matches or beats the tuned Loyola model on both sets (0.957 and 0.903 against medians 0.953 and 0.900). It is essentially *IRLA*:  $LO(G; 0, 0, \gamma) = m + \frac{\gamma}{2} IRLA(G) + O(\gamma^2)$ , the correlation between the two is 0.996 on both sets, and *IRLA* itself comes within 0.003 of the median Loyola model on the nonanes (0.897 against 0.900), so what matters for this property is a term linear in the imbalance, not the tuned exponential. Ridge regression on the full degree-pair count vector predicts the nonanes better than every index model ( $Q^2 = 0.927$  for  $\Delta H_{\text{vap}}$ , 0.682 for  $T_B$ ), consistent with Proposition 15: the imbalance coordinate re-weights degree-pair information rather than adding to it. Protocol details, an exploratory transfer analysis and a post hoc molecule bootstrap (its 95% interval for LO minus GM on nonane  $\Delta H_{\text{vap}}$ ,  $[-0.006, +0.113]$  at budget 200, includes zero) are in Supplementary Section S4.

## 7 Conclusion

The Loyola family extends the Gutman–Milovanović family  $M_{\alpha,\beta}$  [26, 38] by an exponential imbalance factor and reduces to it at  $\gamma = 0$ . Its structural content is exact: the imbalance direction is affinely independent of the two parent directions for  $\Delta \geq 3$  (a class-level statement, Remark 1);  $\log LO$  is globally convex; the pure- $\gamma$  line discriminates exactly the histogram of edge imbalances and loses degree-pair information from tree order 13 (order 16 for  $\Delta \leq 4$ ); two molecular graphs exchange order at most five times along the  $\gamma$  axis; and the bounds of Section 4 extend parent inequalities to  $\gamma \neq 0$ . Generic discrimination is inherited from the parent, and the imbalance coordinate re-weights degree-pair information without adding to it.

Chemometrically, the evidence is diagnostic: every vertex-degree-based

**Table 6.** Nested leave-one-out  $Q^2$  on the octanes (oct.,  $n = 18$ ) and the external nonane set (non.,  $n = 35$  for  $T_B$ , 34 for  $\Delta H_{\text{vap}}$ ). Mean: training-fold mean; Class.: best of the ten classical indices, selected inside each fold; IRLA: one-descriptor model on *IRLA*; Ridge: ridge regression on the edge-degree-pair counts; LO<sub>1</sub>: the fixed, untuned point  $LO(0, 0, 1)$ ; GM, LO: median of  $Q^2$  over seeds 0–99 at budget 200 in the box of half-width 2 (see Table 5 for wider boxes); last column: seeds in which LO beat GM at budgets 200/500. A median of differences, quoted in the text, need not equal the difference of the medians shown. All values are full-precision results rounded once.

| Set, property                | Mean   | Class. | IRLA  | Ridge | LO <sub>1</sub> | GM    | LO    | LO>GM   |
|------------------------------|--------|--------|-------|-------|-----------------|-------|-------|---------|
| oct. $T_B$                   | −0.121 | 0.593  | 0.585 | 0.344 | 0.591           | 0.662 | 0.652 | 53/64   |
| oct. $\Delta H_f$            | −0.121 | 0.746  | 0.593 | 0.654 | 0.639           | 0.831 | 0.689 | 14/24   |
| oct. $\Delta H_{\text{vap}}$ | −0.121 | 0.884  | 0.948 | 0.954 | 0.957           | 0.888 | 0.953 | 100/100 |
| oct. $S$                     | −0.121 | 0.920  | 0.741 | 0.744 | 0.793           | 0.919 | 0.905 | 35/29   |
| oct. $\omega$                | −0.121 | 0.971  | 0.810 | 0.851 | 0.823           | 0.979 | 0.970 | 26/28   |
| non. $T_B$                   | −0.060 | 0.502  | 0.400 | 0.682 | 0.409           | 0.577 | 0.586 | 58/63   |
| non. $\Delta H_{\text{vap}}$ | −0.062 | 0.857  | 0.897 | 0.927 | 0.903           | 0.853 | 0.900 | 95/98   |

index on the octanes operates above an irreducible floor set by two degree-pair collision pairs, and the apparent vaporization-enthalpy gain from freeing  $\gamma$  — seen in every seed of the octane sweep and meeting the pre-registered criterion on the 35 nonanes (mostly compilation values) — disappears once the parent family is searched over a box that does not truncate it (Table 5). Tuning  $\gamma$  thus yields no consistent predictive advantage; the untuned point  $LO(0, 0, 1)$ , essentially *IRLA*, matches the tuned models, and ridge regression on all degree-pair counts predicts the nonanes better than every index model. What the third coordinate offers is parsimony (an exponent near 1 approaches what the parent reaches only at large exponents, as does *IRLA*) and an exact account of what the imbalance direction can and cannot resolve; open problems are listed in Supplementary Section S7.

**Reproducibility.** Data, scripts and a unified verifier that recomputes every tabulated value are at <https://github.com/Singati2/LOYOLA-PAPER>, where the nonane protocol was pre-registered before data collection. The inequalities of Propositions 9 and 10, of Corollary 8 and

Proposition 11, the convexity part of Theorem 3, the rank-4 statement of Proposition 2 for  $\Delta \geq 3$ , and the order-13 profile collision of Section 5.2 have additionally been machine-checked in Lean 4 with Mathlib (graphs modelled by their multisets of edge degree pairs; equality characterizations not formalized); the formalization is in the repository.

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## A Octane isomer property data

For reproducibility, Table 7 lists the full 18-isomer dataset used in Section 6. Units:  $T_B$  in  $^{\circ}\text{C}$ ;  $\Delta H_f$  and  $\Delta H_{\text{vap}}$  in  $\text{kcal mol}^{-1}$  ( $\Delta H_{\text{vap}}$  at 298 K);  $S$  in  $\text{cal mol}^{-1}\text{K}^{-1}$ ;  $\omega$  dimensionless. The  $\Delta H_{\text{vap}}$  values of 3,3-dimethylhexane (8.97), 3-ethyl-3-methylpentane (9.08), 2,2,3-trimethylpentane (8.83) and 2,3,3-trimethylpentane (8.90) are values consistent with the NIST WebBook determinations (see Supplementary Section S1 for the exact-versus-within-uncertainty status of each). The  $\Delta H_{\text{vap}}$  entry of 2,2,3,3-tetramethylbutane is a benchmark-compilation value whose physical reference state is unresolved (Section 6.1).

**Table 7.** Physicochemical properties of the 18 octane isomers.

| Isomer                    | $T_B$ | $\Delta H_f$ | $\Delta H_{\text{vap}}$ | $S$    | $\omega$ |
|---------------------------|-------|--------------|-------------------------|--------|----------|
| <i>n</i> -octane          | 125.6 | −49.82       | 9.92                    | 111.55 | 0.398    |
| 2-methylheptane           | 117.6 | −51.50       | 9.48                    | 109.84 | 0.378    |
| 3-methylheptane           | 118.9 | −50.82       | 9.52                    | 111.26 | 0.371    |
| 4-methylheptane           | 117.7 | −50.69       | 9.48                    | 109.32 | 0.372    |
| 2,2-dimethylhexane        | 106.8 | −53.71       | 8.92                    | 103.13 | 0.339    |
| 2,3-dimethylhexane        | 115.6 | −51.13       | 9.27                    | 108.02 | 0.348    |
| 2,4-dimethylhexane        | 109.4 | −52.44       | 9.03                    | 106.98 | 0.344    |
| 2,5-dimethylhexane        | 109.1 | −53.21       | 9.05                    | 105.72 | 0.357    |
| 3,3-dimethylhexane        | 111.9 | −52.61       | 8.97                    | 104.74 | 0.322    |
| 3,4-dimethylhexane        | 117.7 | −50.91       | 9.32                    | 106.59 | 0.340    |
| 3-ethyl-2-methylpentane   | 115.6 | −50.48       | 9.21                    | 106.06 | 0.330    |
| 3-ethyl-3-methylpentane   | 118.3 | −51.38       | 9.08                    | 101.48 | 0.302    |
| 3-ethylhexane             | 118.5 | −50.40       | 9.48                    | 109.43 | 0.362    |
| 2,2,3-trimethylpentane    | 109.8 | −52.61       | 8.83                    | 101.31 | 0.300    |
| 2,2,4-trimethylpentane    | 99.2  | −53.57       | 8.40                    | 101.81 | 0.305    |
| 2,3,3-trimethylpentane    | 114.8 | −51.73       | 8.90                    | 101.31 | 0.291    |
| 2,3,4-trimethylpentane    | 113.5 | −51.97       | 9.01                    | 102.39 | 0.317    |
| 2,2,3,3-tetramethylbutane | 106.5 | −53.99       | 8.41                    | 93.06  | 0.247    |

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